

MATTER-FIRST AUTOMATED DISCOVERY OF TRANSIENT WARP PRECURSORS

NIKITA M. SHIROKOV*

Independent Researcher, Moscow, Russia

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We build an automated pipeline that searches for warp-precursor spacetimes by proposing matter, solving the Einstein constraints, evolving the spacetime in full 3+1 numerical relativity, and testing whether null rays between fixed asymptotic endpoints arrive earlier than in flat space. MAP-Elites explores a 39-dimensional family of complex-scalar Q-ball shells; **GRTresna** solves the constraints; **GRTeclyn** evolves on GPUs; and a four-dimensional evolving null-geodesic tracer measures the fractional travel-time advantage f_{geo} . The search champion, candidate 118, is a breathing retrograde five-lump shell. In the headline validation run ($L = 128$, $N = 256$, three AMR levels, $t = 30$), light arrives 17.38% earlier than the matched flat path, with all five rays reaching and maximum relative null-Hamiltonian drift 1.69×10^{-3} . Across five completed resolution/domain runs the same configuration shortens null travel by 17.27–19.63% (fine–medium relative difference 5.4%; enlarging $L = 128 \rightarrow 160$ changes the result by 0.36%). The resolution sequence is non-monotonic, so Richardson extrapolation is rejected. Constraint norms stay bounded, but matter confinement falls to about 25% by $t = 30$. The evolution uses an external trajectory controller and does not preserve the initial per-lump phantom signs dynamically. We therefore claim a numerically robust *transient* shortcut in a controlled, driven system—not a conservative Einstein–Klein–Gordon solution or passenger drive.

I. INTRODUCTION

The goal of this work is an *automatic discovery pipeline* for faster-than-light (FTL) warp precursors. We define a warp-precursor as a constraint-satisfying matter configuration that generates a transient, effective null timing advantage prior to its own dispersal: null rays between prescribed asymptotic endpoints traverse the curved spacetime in less coordinate time than the matched flat-space path. Rather than hand-crafting a metric and hoping the implied stress-energy is physical, we want a closed loop that proposes candidates, evolves them under the Einstein equations, and retains only those that pass a gauge-invariant transport test.

The standard route is *geometry-first* (equivalently metric-first): one prescribes $g_{\mu\nu}$, reads off $T_{\mu\nu} = G_{\mu\nu}/8\pi$, and studies frozen slices [3]. That recipe is poorly suited to automated search. The implied matter is rarely the output of a consistent initial-value problem, and under dynamical evolution it typically disperses or collapses, destroying the intended channel. For a quality-diversity optimizer that must evaluate hundreds of candidates, a geometry-first genome is especially hostile: most deformations of $g_{\mu\nu}$ either violate the constraints, have no local matter realization, or fail as soon as time advances—so the archive fills with numerical failures rather than competing mechanisms.

We therefore invert the loop (Fig. 1). The optimizer proposes matter; **GRTresna** returns constraint-compatible initial data; **GRTeclyn** evolves the spacetime; and probes ask whether a null shortcut appears. In this paper “FTL” and “null shortcut” mean the same

operational claim: an *effective* reduction of null travel time relative to flat space between the same asymptotic endpoints—never local superluminal motion outside the light cone. The fractional advantage is f_{geo} [Eq. (7)]; for example $f_{\text{geo}} = 0.17$ means the curved transit is 17% shorter than the flat comparison.

That criterion is load-bearing, not cosmetic. A warp precursor is useful only if it changes what light can do between fixed distant observers. Local coordinate signal speeds, lapse/shift cosmetics, or a positive frozen-slice diagnostic can look “superluminal” while no ray actually connects the endpoints sooner once the metric evolves. Asking whether null geodesics arrive earlier than in flat space is therefore the operational definition of the search target: a gauge-invariant transport certificate that the optimizer must earn, not a post-hoc plot.

Equally important is the *scoring metric* that turns each evolution into the fitness $S(\theta)$ used by the archive [Eq. (3)]. MAP-Elites converges to whatever S rewards. In earlier campaigns with weaker objectives the search reliably *cheated*: it harvested coordinate proxies while matter dispersed (“disperse-then-warp”), banked frozen-slice positives that failed the evolving geodesic test, or climbed scores through numerical pathologies—early horizons, integrator crashes, and other simulation-failure modes that accidentally produced large proxy signals. The present objective therefore rewards only the evolving 4D shortcut and persistence, gates health terms by nontriviality, penalizes exotic excess and horizon formation, and dispersion-gates transport credit so a flying-apart cloud cannot buy a high score. Without that metric design the archive does not discover physics; it discovers loopholes.

The search engine is MAP-Elites [7], a quality-diversity algorithm that keeps one elite per cell of a behavior archive rather than a single global optimum. Here the archive axes are peak shortcut strength and FTL life-

* shirokov.nm@phystech.edu

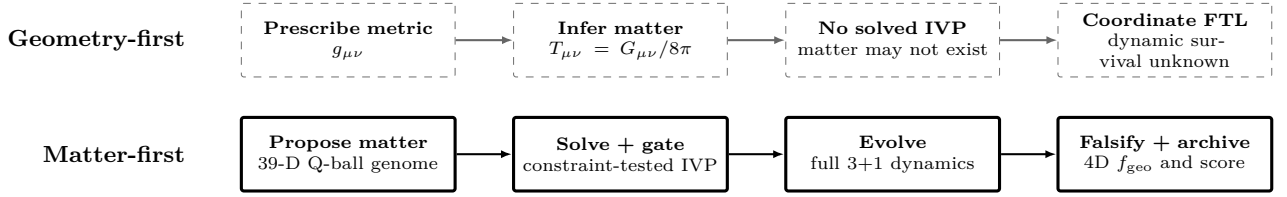


FIG. 1. Geometry-first versus matter-first discovery. Dashed: prescribe a metric and infer its stress-energy. Solid: propose matter, solve and test an initial-value problem (IVP), evolve it, and archive only candidates that pass the 4D null-transport test. Candidate 118 retains the source-model limitations described in Sec. II E.

time, so distinct transport mechanisms can coexist—but only among candidates that already cleared the scored falsification gate. We prefer this over a pure CMA-ES hill-climb, which collapses onto one basin and discards diverse near-misses, and over reinforcement learning, which would require dense credit assignment across expensive GPU evolutions with sparse, delayed rewards. After MAP-Elites identifies a high-scoring basin, CMA-ES is the mandatory local refiner that tunes the champion genome before any high-resolution replay or validation matrix; the discovery stage itself remains deliberately diversity-preserving.

A. Methodological Rigor

Three pillars distinguish this pipeline from geometry-first studies:

1. **Initial value problem:** constraint-clean data from physical scalar sources, not coordinate scripts.
2. **Dynamic feedback:** full BSSN/CCZ4 evolution; dispersing matter collapses the channel.
3. **4D geodesic falsification:** the headline diagnostic is `ftl_geo_evolving`—null rays integrated through the *time-evolving* metric stack. Frozen-slice geodesics are retained as diagnostics but are not the transport certificate.

B. Contributions

- (i) An end-to-end matter-first discovery pipeline—MAP-Elites proposal, CMA-ES local refinement of the archive champion, constraint solve, GPU evolution, and matched evolving-geodesic scoring—with an explicit reward rule applied to every evaluation (Secs. II–III).
- (ii) A hierarchical campaign protocol: quality-diversity search, CMA-ES hill-climb on the top genome, then longer higher-resolution replay and a resolution/domain matrix only on the refined champion (Sec. V).

- (iii) Explicit numerical limits of the surviving candidates: non-monotonic resolution behavior, dispersing exotic matter, and preliminary finite-radius curvature extraction—not passenger transport (Sec. VI).

II. MATTER-FIRST PIPELINE

A. Two-code toolchain

Two open numerical-relativity codes carry the loop. **GRTresna** is a constraint solver: given a matter configuration it returns constraint-satisfying initial data $(\gamma_{ij}, K_{ij}, \phi, \Pi)$ on the initial slice, so that the Hamiltonian and momentum constraints hold before any evolution. **GRTeclyn** [8, 9] is a GPU-accelerated, AMReX-based code that evolves those data forward with the CCZ4 formulation of the Einstein equations under moving-puncture gauge. Neither code is told what metric to produce; the geometry is whatever the proposed matter sources.

B. Search Loop

One evaluation maps a parameter vector θ to a scalar fitness: $\theta \rightarrow \text{initial data} \rightarrow \text{GRTeclyn} \rightarrow \text{diagnostics} \rightarrow S(\theta)$. The optimizer proposes θ , **GRTresna** solves the constraints, **GRTeclyn** evolves the spacetime on GPU, and post-processing probes reduce the evolution to the fitness $S(\theta)$ that steers the next proposal.

The matter is a canonical complex scalar field ϕ with conjugate momentum Π and potential $V(\phi)$. In the 3+1 split, a scalar field contributes an energy density ρ and a momentum density S_i that act as the sources in the Hamiltonian and momentum constraints,

$$\rho = \frac{1}{2}\psi^{-4}|\tilde{\nabla}\phi|^2 + \frac{1}{2}\Pi^2 + V(\phi), \quad S_i = -\Pi\partial_i\phi, \quad (1)$$

where ψ is the conformal factor and $\tilde{\nabla}$ the conformal spatial gradient. The three terms in ρ are, in order, the gradient (spatial shape), kinetic (time variation), and potential (rest) energy of the field; S_i is nonzero only when the field oscillates in time ($\Pi \neq 0$) with a spatial gradient, which is what lets a moving scalar lump carry momentum

and drive a shift. Equation (1) is therefore the precise statement of “the matter tells the geometry what to be”: θ sets ϕ and Π , and **GRTresna** returns the (γ_{ij}, K_{ij}) consistent with these sources.

The word “canonical” describes the Klein–Gordon equation of motion, not the sign with which every initial lump gravitates. For each lump the optimizer stores $x_{\text{ex},k} \in [0, 1]$, but configuration construction rounds it to $e_k = \text{round}(x_{\text{ex},k})$ and sets $s_k = (-1)^{e_k}$. Thus there is no physical fractional interpolation. In the **GRTresna** constraint solve the signed independent-lump approximation is

$$\rho_{\text{ID}} = \sum_k s_k \rho_k, \quad (S_i)_{\text{ID}} = - \sum_k s_k \sum_{A=1}^2 \Pi_{A,k} \partial_i \phi_{A,k}, \quad (2)$$

with inter-lump stress-tensor cross terms omitted. The profiles themselves are then superposed into one painted complex field $\Phi = \phi_1 + i\phi_2$. Candidate 118 rounds to $s_k = (+1, -1, -1, -1, -1)$: one canonical and four phantom initial-data sources.

C. Q-ball building blocks

The matter is built from Q-balls, and this choice was forced on us by failure. Earlier ansatz families—spherical-harmonic scalar shells and free massive (Klein–Gordon) wave packets—have no bound state and simply spread: their structural persistence collapses to a few percent well before the end of the evolution, taking any transient shortcut with them (Sec. VIA). A Q-ball is a localized, non-topological soliton of a complex scalar field: the field carries a conserved Noether charge under a global phase symmetry, and that charge holds a compact lump of field together against dispersal. This self-binding is precisely what the earlier families lacked. We choose a *complex* rather than real scalar for two reasons. First, only a complex field has the phase symmetry that supports a stable charged lump. Second, its harmonic phase rotation $\phi \propto e^{-i\omega t}$ makes $\Pi = \dot{\phi} \neq 0$, so by Eq. (1) the lump carries a momentum density S_i and can source a nontrivial shift—the ingredient a transport channel needs. Each lump starts from a spherically symmetric equilibrium profile obtained by solving the radial ODE for the chosen sextic potential ($m = 1$, $\lambda = 640$, $\mu = 85333$, $\omega = 0.8$); these fixed microphysical constants are not searched.

D. Shell Parameterization

The search vector has 39 dimensions because the arrangement of the lumps, not the lump microphysics, is what is optimized. Five lumps each carry seven searchable placement parameters—orbital radius R_0 , phase, two tilt angles, radial velocity, angular (rotation) velocity, and a source-sign selector—giving $5 \times 7 = 35$ per-lump dimensions. Four collective variables set the shell’s breath-

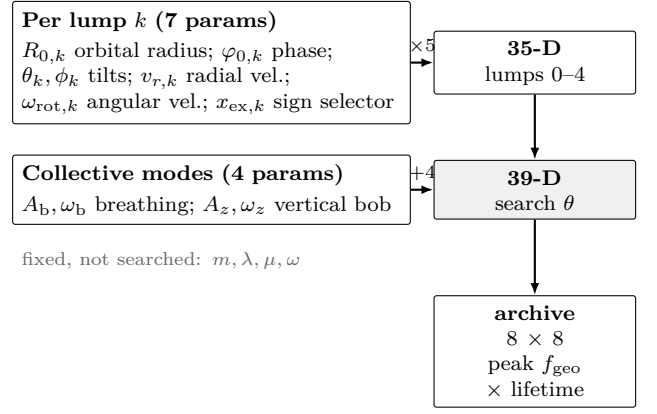


FIG. 2. Search parameterization of the trajectory Q-ball shell: $5 \times 7 = 35$ per-lump placement parameters plus 4 collective modes give the 39-D genome θ . MAP-Elites stores one elite per cell of an 8×8 archive indexed by peak shortcut and FTL lifetime. Q-ball microphysics is held fixed.

ing amplitude and frequency and its vertical oscillation amplitude and frequency, for $35 + 4 = 39$ in total (Fig. 2). Although sampled continuously, the sign variable is applied as the binary selector above; it lets the optimizer choose which initial-data lumps source positive or negative stress-energy. MAP-Elites [7] keeps one elite per cell of an 8×8 behavior archive indexed by peak shortcut strength and FTL lifetime, which preserves distinct mechanism families rather than collapsing onto a single optimum.

E. Constraint gate and source-model limitation

A post-load constraint gate rejects inconsistent initial data before any GPU time is spent: the **GRTresna** Hamiltonian and momentum residuals must fall below 5%, otherwise the candidate is discarded and logged. This is stronger than evolving a hand-drawn metric, but the audited candidate has an important source-model mismatch. Its **GRTeclyn** replay uses `gtrtesna_complex_scalar`: one summed complex field with global canonical sign +1. The four per-lump phantom signs in Eq. (2) affect the initial constraint solve but are not retained in the evolution stress tensor. A two-field canonical/phantom implementation exists in the codebase but was not used for this campaign. Consequently, passing the post-load gate and maintaining bounded CCZ4 constraints establish numerical control, not an exactly matched conservative matter model.

F. Scoring: how every evaluation is rewarded

Every evaluation in the campaign is reduced to one fitness value $S(\theta)$ by the same fixed rule, so candidates are ranked consistently. Each named quantity below is

TABLE I. Reward terms in the campaign objective [Eq. (3)], applied identically to every MAP-Elites evaluation.

symbol	rewards / penalizes	weight
$f_{\text{geo}}^{\text{evol}}$	4D evolving null shortcut	1000
$f_{\text{geo}}^{\text{op}}$	geodesic (frozen) shortcut	1000
P_{ftl}	FTL persistence (gated)	600
F_{op}	operational coord. FTL (gated)	200
C_{curv}	curvature activity	200
G_{nt}	nontrivial geometry	5
surv	structural survival	$150 g_h$
E	energy-condition health	$60 g_h$
stab	metric stability	$40 g_h$
cmv	co-moving stability	$30 g_h$
I	instability penalty	$15 g_h$
$\mathcal{H}_{\text{ok}}$	constraint health	$10 g_h$
X_{exotic}	exotic-matter penalty	$40 w_x$
H_{hor}	horizon-formation penalty	200

a normalized diagnostic in $[0, 1]$ (penalties are negative); the score is their weighted sum,

$$S = 1000 f_{\text{geo}}^{\text{evol}} + 1000 f_{\text{geo}}^{\text{op}} + 600 P_{\text{ftl}} + 200 F_{\text{op}} + 200 C_{\text{curv}} + 5 G_{\text{nt}} + g_h \Sigma_{\text{health}} + 40 w_x X_{\text{exotic}} + 200 H_{\text{hor}}, \quad (3)$$

with the health block

$$\Sigma_{\text{health}} = 150 \text{surv} + 60 E + 40 \text{stab} + 30 \text{cmv} + 10 \mathcal{H}_{\text{ok}} + 15 I, \quad (4)$$

gated by the nontriviality factor $g_h \in [0, 1]$ so that a trivial (flat) configuration earns no health reward. Table I lists what each term measures. Two design choices matter. First, transport dominates: the evolving 4D shortcut $f_{\text{geo}}^{\text{evol}}$ and its geodesic counterpart carry the largest weights, so the score chiefly tracks genuine null-transport gain. Second, the persistence and operational terms $P_{\text{ftl}}, F_{\text{op}}$ are *dispersion-gated*—multiplied by the fraction of matter still confined—so a candidate cannot bank a co-ordinate superluminal channel after its lumps have flown apart. The exotic penalty $X_{\text{exotic}} < 0$ charges for energy-condition violation at campaign weight w_x . All single scores quoted later are values of this S on a $t = 16$ evolution: search rankings, not physical observables, and never compared across runs of different duration.

For candidate 118 the reward is dominated by the evolving-shortcut term ($f_{\text{geo}}^{\text{evol}} \approx 0.31$; the frozen geodesic shortcut $f_{\text{geo}}^{\text{op}}$ was zero at $t = 16$), the gated persistence term ($P_{\text{ftl}} = 0.35$), and curvature activity ($C_{\text{curv}} = 0.61$), with healthy constraint growth (0.95) and full survival, against an exotic-matter penalty ($X_{\text{exotic}} = -1.6$) and a mild instability penalty. Its total is $S = 603.39$.

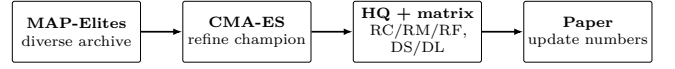


FIG. 3. Outer campaign ladder. High-resolution replay and the resolution/domain matrix run only on the CMA-ES-refined champion, not on a raw MAP-Elites elite.

G. Search and validation stages

- **Stage 1 — MAP-Elites discovery:** quality-diversity search at $L = 64$, $N = 128$, one refinement level, and $t = 16$ (bicomplex matter; per-lump gravitational signs retained in evolution). Produces a diverse archive and a provisional score leader.
- **Stage 2 — CMA-ES refinement:** warm-start from the MAP-Elites champion (and optionally its nearest elites); local covariance-matrix adaptation at the *same* search resolution and scoring rule. Only the refined top genome is promoted further.
- **Stage 3 — production replay (HQ):** the frozen CMA-ES champion is re-solved and evolved at higher resolution / longer t_{stop} (e.g. $L = 128$, $N = 256$, three refinement levels), using multiple emission times and multi-ray geodesic probes.
- **Stage 4 — numerical matrix:** matched coarse/medium/fine resolution runs and small/medium/large domains test whether the shortcut survives discretization and boundary changes.
- **Stage 5 — waveform run:** after the matrix gate passes, a dedicated longer run samples Ψ_4 more densely at several extraction radii.

Note. Earlier spiral_v2 / candidate 118 numbers are quarantined (ID/evolution sign mismatch); paper tables will be rewritten from the bicomplex MAP-Elites $\rightarrow$ CMA-ES champion once Stages 1–2 complete.

III. SEARCH METHODOLOGY

A. MAP-Elites quality-diversity

MAP-Elites [7] is a quality-diversity search: instead of returning one champion, it maintains an 8×8 archive with at most one elite per behavior cell. A candidate displaces the occupant of its cell only if its score $S(\theta)$ [Eq. (3)] is higher. The two archive axes are peak shortcut strength and FTL lifetime, so the campaign can keep short-lived strong channels and longer weaker ones side by side. The bicomplex campaign (qball_traj_bicomplex_v1) scheduled 200 evaluations: 148 completed GPU evolution, 50 failed the post-load gate, one failed the GRTresna admission gate, and one failed the constraint solve. Nine cells were occupied

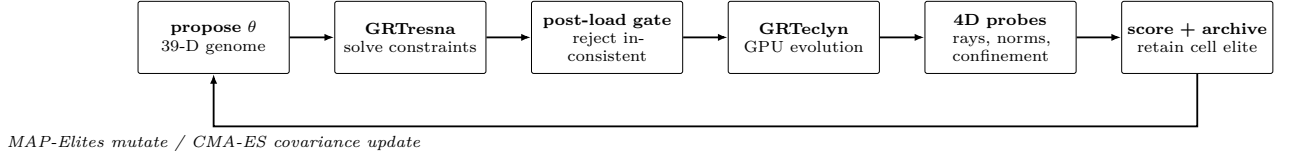


FIG. 4. Inner evaluation loop shared by MAP-Elites and CMA-ES. Failed constraint solves and post-load checks never enter GPU evolution; successful evaluations compete for archive cells (MAP-Elites) or update the CMA distribution.

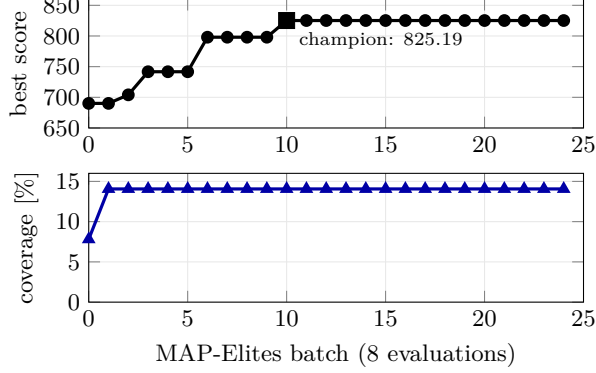


FIG. 5. MAP-Elites search progress over 200 evaluations (qball_traj_bicomplex_v1). Top: running best archive score. Bottom: fraction of occupied behavior cells. The final champion (candidate 87) enters cell (7,7) at batch 10 with score 825.19. Curves are read from the stored campaign history.

(14.06% coverage). All nontrivial elites lie in the highest lifetime row, while peak strength spans the archive. The best score rose from 690.1 to 825.19 and last improved at batch 10, when candidate 87 entered cell (7,7).

B. Archive outcome and promotion

The bicomplex winner (candidate 87, $S = 825.19$) combines a search-tier evolving shortcut of $f_{\text{geo}} = 25.3\%$ with operational score 0.396, constraint-growth score 0.983, and confinement retention 0.396 (dispersion-gated). Its five radial scales span $R_0 = 2.53$ – 5.64 ; three lumps are phantom and two are canonical (recipe_scalar_field_signs = +1, −1, −1, +1, −1). All angular velocities are retrograde and slow ($|\omega_{\text{rot}}| \lesssim 0.05$). CMA-ES refinement of this genome is the next stage; HQ and the resolution/domain matrix run only after that local hill-climb freezes the champion.

Search-tier confinement already falls to $\sim 40\%$ by $t = 16$, so longer production runs are expected to stress structural persistence even if the evolving null shortcut remains positive.

TABLE II. Provisional MAP-Elites hand-off for candidate 87 (search tier only). HQ / matrix rows will be filled after CMA-ES refinement.

stage	(L, N, t_{stop})	rays	peak f_{geo}
MAP-Elites	(64, 128, 16)	3/3	25.35%
CMA-ES refine	—	—	(in progress)
HQ / matrix	—	—	(pending)

IV. DIAGNOSTICS AND PROBE VALIDATION

A. The 4D evolving null-geodesic trace

The transport claim rests on tracing light rays through the simulated spacetime, not on reading coordinate speeds off a slice. A coordinate operational shortcut F_{op} [Eq. (6)] can flag tilted light cones, but it is partly gauge dependent and can be inflated by a coordinate choice. The authoritative probe therefore integrates the null-geodesic equations

$$\dot{x}^\mu = g^{\mu\nu} k_\nu, \quad \dot{k}_\mu = -\frac{1}{2}(\partial_\mu g^{\alpha\beta}) k_\alpha k_\beta, \quad (5)$$

for a photon with position x^μ and momentum k_μ . “4D” means the ray is not confined to one snapshot: it is pushed through a stack of stored time slices, so at each step the metric $g_{\mu\nu}(t, \mathbf{x})$ used is the one that actually exists when the photon is at that point. This distinguishes it from a *frozen* trace, which wrongly assumes a single slice persists for the whole flight.

The construction is as follows. A ray is emitted at time t_{emit} from a fixed source point toward a fixed receiver. The quantities that enter the shortcut formulas are *transit durations*, not absolute grid arrival times: t_B^{curved} (respectively t_{min}) is the coordinate time elapsed along the curved path after emission, and t_B^{flat} (respectively t_{flat}) is the matched flat-space flight duration between the same endpoints. Absolute arrival on the grid is $t_{\text{arrival}} = t_{\text{emit}} + t_B^{\text{curved}}$. The fractional advantage is

$$F_{\text{op}} = \frac{t_B^{\text{flat}} - t_B^{\text{curved}}}{t_B^{\text{flat}}}, \quad (6)$$

$$f_{\text{geo}} = \max\left(0, \frac{t_{\text{flat}} - t_{\text{min}}}{t_{\text{flat}}}\right), \quad (7)$$

so $f_{\text{geo}} = 0$ means no shortcut and $f_{\text{geo}} = 0.17$ means the curved transit is 17% shorter than the flat comparison.

A launch is accepted only if its rays stay physical: the relative drift in the null Hamiltonian $g^{\mu\nu}k_\mu k_\nu = 0$ must remain below 10^{-2} , which rejects rays that have accumulated interpolation error. The simulation itself runs to $t = 30$, but that is the evolution horizon of the metric stack, not the emission schedule. A ray launched at t_{emit} still needs a long remaining segment of evolved geometry to reach the far endpoint (flat transit is $t_{\text{flat}} = 14.4$), so late launches near $t = 30$ cannot finish a fair comparison. We therefore sweep seven early launch times, $t_{\text{emit}} = 0, 2, \dots, 12$ (five rays each), and report the largest accepted f_{geo} —the best moment to “surf” the channel. Across the validation matrix the peak sits at $t_{\text{emit}} = 8$ and has already fallen by $t_{\text{emit}} = 12$, so the window covers the opening of the channel rather than cutting it off.

B. Probe controls

Prescribed Alcubierre [4] ($f_{\text{geo}} \approx 32\%$) and Minkowski ($f_{\text{geo}} = 0$) calibrate the codebase. A previously identified negative control, `eval086`, has a positive frozen-slice diagnostic but fails the evolving HQ trace. Candidate 118 supplies the complementary positive case: every accepted launch in all five validation runs has 5/5 rays reaching and passes the null-Hamiltonian gate.

V. RESULTS: CANDIDATE-118 WARP PRECURSOR

A. Provenance and genome

Candidate 118 is the score leader of the completed MAP-Elites campaign. Its provenance freezes the exact 39-dimensional genome, source checksums, solver parameters, and source commit. The configuration is not a fitted metric: it is a dynamic five-lump scalar shell whose matter currents source the solved conformal geometry and shift. The present validation matrix was run with `GRTECLYN_FRAMES=0`; consequently, no field-map panels from these runs are presented.

B. Matter dynamics

The five Q-balls do not share one rigid orbit. Lump k moves in its own tilted orbital plane, with the support center prescribed by

$$\begin{aligned} r_k(t) &= \max[r_{\min}, R_{0,k} + v_{r,k}t + A_b \sin(\omega_b t)], \\ \varphi_k(t) &= \varphi_{0,k} + \omega_{\text{rot},k}t, \quad z_b(t) = A_z \sin(\omega_z t), \end{aligned} \quad (8)$$

followed by rotation into the laboratory frame through the two tilt angles. Thus each lump spirals radially, rotates in a different plane, and shares a collective radial “breathing” and vertical oscillation. For candidate 118, $A_b = 1.60$, $\omega_b = 2.67$, $A_z =$

1.50, and $\omega_z = 1.51$. The initial radii are $R_0 = (6.64, 7.52, 1.96, 3.88, 4.86)$; all angular velocities are negative, $\omega_{\text{rot}} = (-0.045, -0.040, -0.150, -0.040, -0.061)$, so all five trajectories are retrograde. The innermost lump therefore turns fastest. Radial drifts are small [$v_r = (0.007, 0.003, 0.063, 0.069, -0.037)$], while the shared breathing mode produces most of the short-time radial motion. Four lumps are predominantly phantom in the initial-data source; lump 0 is canonical after rounding.

Figure 6 shows these prescribed support-center trajectories at four times. The Q-ball fields are initially boosted along the corresponding motion. During evolution, a localized proportional-derivative (PD) controller drives the two real components of the complex field toward a moving target profile:

$$\begin{aligned} \partial_t \Pi_A|_{\text{PD}} &= -wk_p(\phi_A - \phi_A^*) \\ &\quad - wk_d(\Pi_A - \Pi_A^*), \quad A = 1, 2, \end{aligned} \quad (9)$$

with $k_p = 12$, $k_d = 7$, and w the product of a localized envelope and a constraint-sensitive governor. This is the precise meaning of the “moving well.” It is an external, non-Lagrangian source in the scalar momentum equations; no corresponding pump stress-energy is added to $T_{\mu\nu}$. Therefore $\nabla_\mu T^{\mu\nu} = 0$ is not exact and the contracted Bianchi identity is not satisfied by a closed matter system.

The RadialRecipe path used here records no pump-work integral, and the search score does not penalize net pump work; its exotic and stability penalties are different quantities. The bounded constraints ($\|\mathcal{H}\|_2 \leq 5.31 \times 10^{-3}$, $\|\mathcal{M}\|_2 \leq 7.48 \times 10^{-4}$ in RM) show that the forcing remains numerically controlled, not that the evolution is a conservative Einstein–Klein–Gordon solution. The support centers are the intended dynamical scaffold, not tracked density isosurfaces. The field follows them imperfectly: confinement falls to 25.6% and rms spread grows by 2.76 by $t = 30$. Candidate 118 is therefore a transient, externally driven configuration, not a bound five-body soliton.

C. 4D falsification certificate

The headline claim is operational: in the RM run, null rays between the fixed endpoints arrive earlier than in flat space. The evolving-geodesic artifact reports:

peak 4D f_{geo}	17.38%
rays reached	5/5
t_{emit}	8.00
$t_{\text{transit}}^{\text{curved}}$	11.897 ($= t_{\text{arrival}} - t_{\text{emit}}$)
$t_{\text{transit}}^{\text{flat}}$	14.40
$h_{\text{quality_ok}}$	true ($h_{\text{rel}}^{\text{max}} = 1.69 \times 10^{-3}$)
search-tier peak	17.65% (3/3 rays)

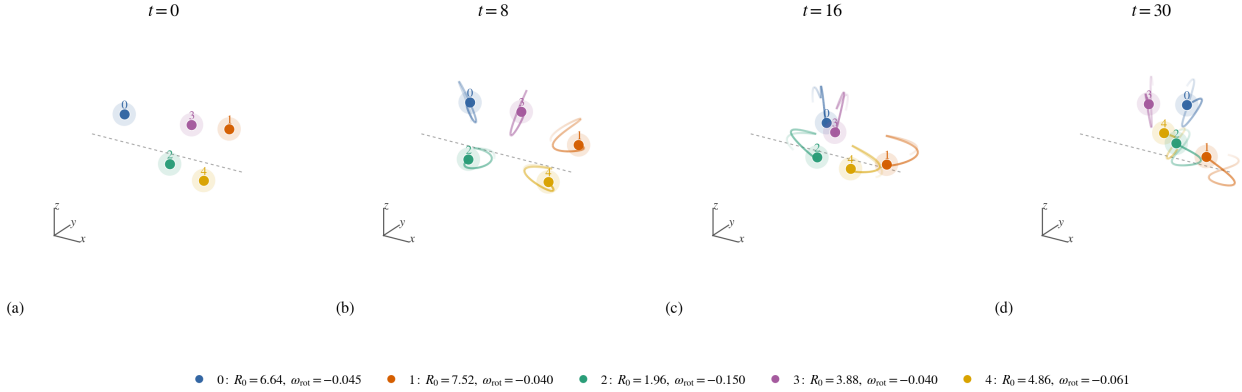


FIG. 6. Three-dimensional dynamics encoded by the frozen candidate-118 genome. Panels show the five prescribed PD-controller centers at $t = 0, 8, 16$, and 30 ; short fading trails cover the preceding four code-time units and indicate the direction of motion. Colors and labels identify the lumps, and the dashed line marks the x -directed null-probe axis. The translucent halos schematically indicate the compact support scale and are not density isosurfaces. These paths are evaluated with exactly the trajectory law used by `GRTeclyn` [Eqs. (8) and (9)]; they are not freely evolved matter trajectories.

TABLE III. Completed candidate-118 validation matrix. All runs use three AMR levels and stop at $t = 30$. RM is the shared headline baseline.

run	meaning	L	N	h	peak f_{geo}
RC	resolution coarse	128	192	0.667	19.63%
RM	resolution medium	128	256	0.500	17.38%
RF	resolution fine	128	384	0.333	18.38%
DS	domain small	96	192	0.500	17.27%
DL	domain large	160	320	0.500	17.45%

Explicitly, $(t_{\text{transit}}^{\text{flat}} - t_{\text{transit}}^{\text{curved}})/t_{\text{transit}}^{\text{flat}} = (14.40 - 11.897)/14.40 = 17.38\%$, matching Eq. (7). Absolute arrival $t_{\text{arrival}} = 19.897$ is larger than t_{flat} only because emission is delayed; the curved *duration* is shorter. The production value differs from the search value by 1.5% relative. The endpoint timing corresponds to an effective $c/(1 - f_{\text{geo}}) \simeq 1.21c$ relative to the matched flat path; locally the ray remains null.

D. Resolution and domain matrix

The five-run matrix splits into a resolution ladder at fixed domain $L = 128$ —coarse RC ($N = 192$), medium RM ($N = 256$), fine RF ($N = 384$)—and a domain ladder at fixed spacing $h = 0.5$ —small DS ($L = 96$), medium RM ($L = 128$), large DL ($L = 160$). RM is the shared headline baseline.

Across the matrix, the same genome shortens null travel by 17.27–19.63% relative to flat space. RF and RM differ by 5.4% relative, below the preregistered 10% matrix tolerance. DL and RM differ by 0.36%; even DS differs from RM by only 0.63%. Boundary placement is therefore subdominant over $L = 96$ –160. For the do-

main used here, the $t = 30$ stop precedes the arrival of boundary-reflected gauge modes at the central region and the extraction spheres: in $1 + \log$ slicing those modes propagate at characteristic speeds up to $\sim \sqrt{2}c$, so a one-way crossing from the outer boundary at distance $L/2$ requires $t \gtrsim L/(2\sqrt{2})$ ($\approx 34, 45$, and 57 for $L = 96, 128$, and 160). The f_{geo} peak at $t_{\text{emit}} = 8$ with arrival near $t \approx 20$ is therefore causally interior to the domain.

For unequal spacings, a Richardson estimate would assume

$$Q(h) = Q_0 + Ch^p, \quad \frac{Q_c - Q_m}{Q_m - Q_f} = \frac{h_c^p - h_m^p}{h_m^p - h_f^p}. \quad (10)$$

The sequence 19.63% $\rightarrow$ 17.38% $\rightarrow$ 18.38% changes sign between successive differences. The extrapolation is therefore rejected as non-monotonic: no observed order, continuum value, or GCI is quoted. This is not evidence of a buggy integrator. Because the Q-ball matter is actively dispersing, the physical volume that triggers the gradient tag (`regrid_threshold`=0.02) changes dynamically with resolution. Consequently $N = 384$ does not merely slice the $N = 256$ grid finer: it alters the total volume covered by level-3 refinement. In dynamic, dispersing spacetimes with threshold-based AMR, strict asymptotic monotonic convergence is frequently lost due to varying fine-grid coverage. Finite-resolution agreement across the matrix supports robustness, not formal continuum extrapolation.

E. Evolution constraints

For RM, the volume-weighted norms remain below $\|\mathcal{H}\|_2 = 5.31 \times 10^{-3}$ and $\|\mathcal{M}\|_2 = 7.48 \times 10^{-4}$ through $t = 30$ (Fig. 8). Across RC, RM, and RF the Hamiltonian maxima are 5.335, 5.313, 5.317×10^{-3} and the momentum maxima are 7.497, 7.478, 7.478×10^{-4} . They are

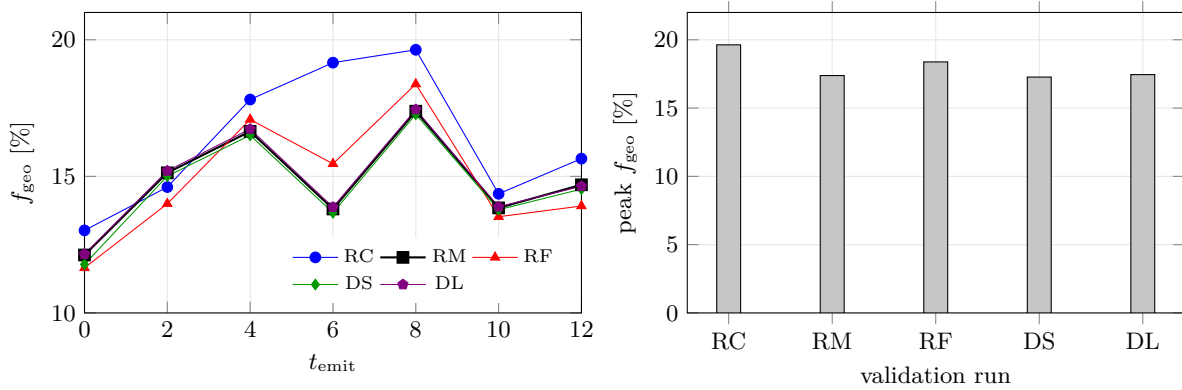


FIG. 7. **Left:** evolving null-geodesic emission sweeps ($t_{\text{emit}} = 0, 2, \dots, 12$) for all five validation runs; the metric evolves to $t = 30$, but late launches lack enough remaining geometry for a complete transit. **Right:** accepted peak in each run (all at $t_{\text{emit}} = 8$). Every point shown has 5/5 rays reaching and passes the null-Hamiltonian quality gate.

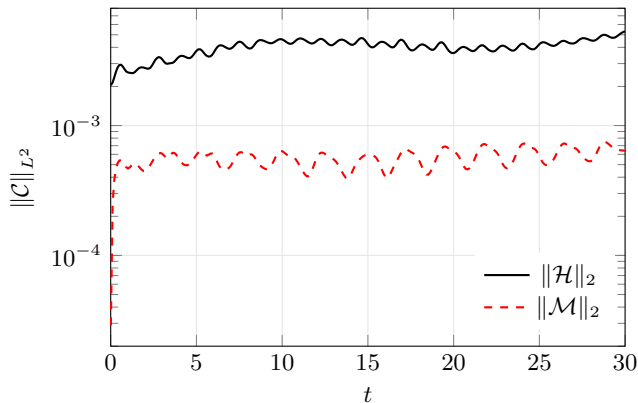


FIG. 8. Constraint norms in the headline RM run. The complete 3000-step diagnostic is plotted from a decimated table for typesetting only; maxima are evaluated from the full data.

bounded but non-monotonic or plateaued, so Richardson extrapolation is not claimed for the constraints either.

F. Frozen diagnostics, confinement, and curvature extraction

RM's frozen-slice geodesic reaches 19.72%, compared with the evolving sweep peak of 17.38%. The distinction is physical: a frozen trace assumes that one snapshot persists for the entire ray flight, whereas the 4D trace samples the actual changing metric. Only the latter is used for the transport claim. The frozen f_{geo} time series itself peaks later, at $t \simeq 20.16$ with 24.6% (GW metric stack; RM agrees to 0.1%), after the best evolving launch time $t_{\text{emit}} = 8$.

The matter structure is not durable. In RM, structural persistence and final confinement are 25.6%, while the rms spread grows by a factor 2.76. The five matrix runs end with 23–26% persistence. Thus successful ray transport coexists with substantial dispersal.

G. Descriptive peak-metric fit

A natural follow-up to the original search proposal is to ask whether the geometry at peak shortcut admits a simple closed form, in the spirit of an Alcubierre shape function. We attempt only a *descriptive* fit. From the GW metric-stack cache at $t = 20.16$ we reconstruct the ADM lapse α , shift component β^x , and conformal factor $\chi = (\det \gamma)^{-1/3}$ on the equatorial midplane, then fit degree-4 polynomials in cylindrical radius r to the azimuthally binned means (Fig. 9). The binned curves are smooth ($R^2 \approx 0.95$ – 0.999), but this is not an analytic spacetime: angular scatter in α is $\sim 72\%$ of its variance, so the configuration is strongly non-axisymmetric, and a pure r -dependent model cannot represent the five-lump shell. Along the probe x -axis alone the shift fit drops to $R^2 \approx 0.48$. We therefore report the polynomials as a local cartoon of the peak slice, not as a reproducible fundamental metric. This demonstrates a fundamental limitation of geometry-first literature. Analytic templates such as Alcubierre [4] or Natário [5] inherently assume clean, symmetric metrics. Dynamically evolved, matter-sourced null shortcuts rely instead on highly asymmetric, messy metric configurations that cannot be captured by standard axisymmetric shape functions.

The post-processed spherical extraction also contains an $\ell = 2, m = 0$ curvature mode at $R_{\text{ext}} = 12, 18, 24$, with peak $|r\Psi_4^{2,0}| \simeq 2.05 \times 10^{-2}$ at the innermost radius. The RM record has only 22 samples, so Fig. 10 is a pipeline verification, not a spectral or detector result. Because the primary f_{geo} peak occurs during a phase heavily contaminated by initial-data relaxation and dynamic matter dispersal, and the extraction cadence is sparse, we defer quantitative waveform claims, no-pump ablation studies, and QNM fitting to future work featuring wave-zone characteristic extraction (CCE).

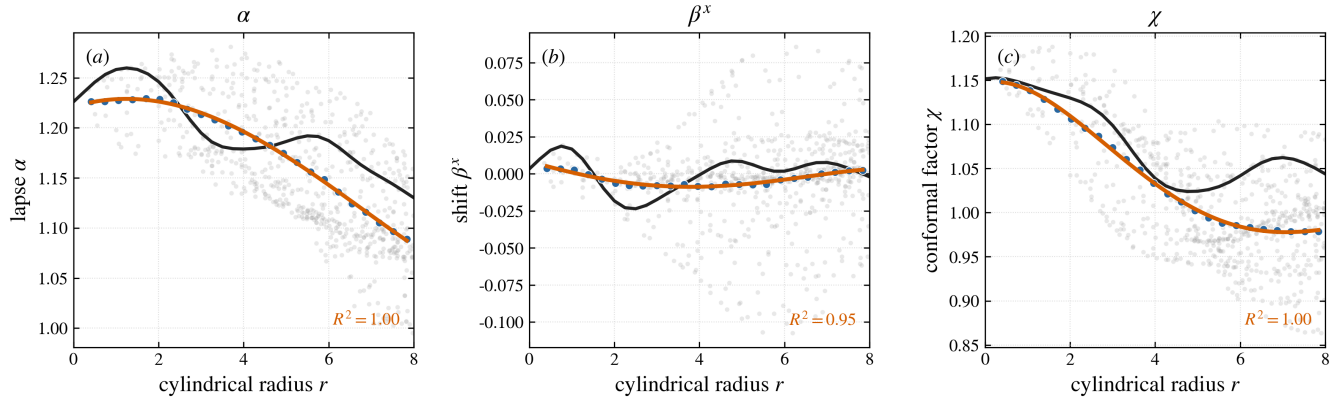


FIG. 9. Descriptive ADM profiles at the frozen-shortcut peak ($t = 20.16$). Grey points: equatorial midplane samples. Black: x -axis cut through the probe line. Blue: azimuthal r -bins. Orange: degree-4 polynomial fit to the binned means. High R^2 on the means coexists with large angular scatter, so no axisymmetric closed form reproduces the metric.

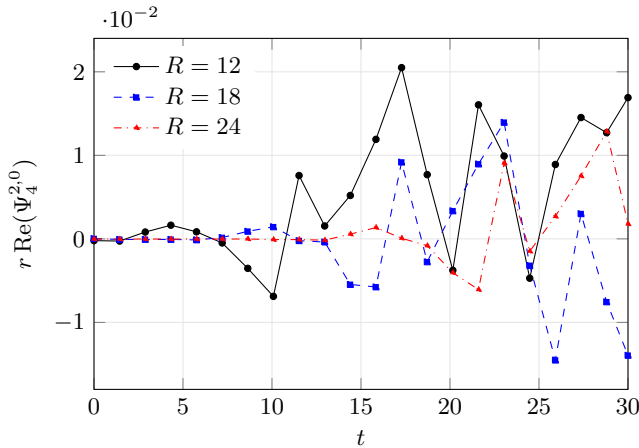


FIG. 10. Preliminary post-processed curvature mode from RM. Sparse cadence and near/intermediate extraction radii preclude a quantitative waveform claim.

H. What moves faster than light

Nothing physical moves outside its local light cone. The quoted 17–20% is a *null-geodesic timing advantage* through the evolved geometry relative to a flat reference path between the same endpoints. Coordinate light-speed maps visualize the tilted cones but do not establish the result by themselves.

VI. DISCUSSION

A. The dispersion problem

Candidate 118 should be read against the many configurations that did not work. Across roughly a dozen ansatz families and several full MAP-Elites campaigns we repeatedly found the same failure mode: the supporting matter disperses and the transport channel closes with

it. Spherical-harmonic shells yielded a single weak hit in ~ 200 evaluations. Free massive scalar packets have no bound state and blow apart almost immediately. Bicomplex phantom lumps opened a gauge-invariant shortcut but lost $\sim 98\%$ of their structure by $t = 16$ (structural persistence ≈ 0.02). Boosted and stiffened Q-ball variants improved early confinement but still sheared apart once placed on an orbit. This is why the score explicitly gates transport by confinement (Sec. II F): without it, the optimizer is rewarded for a coordinate superluminal channel produced by matter that has already flown apart. Candidate 118 is the most persistent genome found, yet even it retains only about a quarter of its confined matter by $t = 30$. The honest summary is that a *transient* shortcut is reproducible and robust, while a *long-lived*, well-confined channel remains an open problem.

B. Interpretation

Candidate 118 demonstrates that matter-first search with 4D falsification can identify a constraint-gated, externally driven evolution whose transport gain survives production resolution, further refinement, and a larger domain. The strongest claim is finite-resolution robustness of the measured null timing: all five runs are positive and all ray-quality gates pass. The data do not support a universal convergence order. They also do not establish a closed Einstein–Klein–Gordon solution: the initial solve uses four phantom-signed lump sources, the evolution uses one globally canonical field, and the PD controller supplies untracked external work.

C. Applications and next steps

The pipeline supports (i) inverse design of null transport under energy-condition penalties, (ii) QD atlases of mechanism families, and (iii) Pareto maps of shortcut ver-

sus exotic cost and persistence. Immediate priorities are to add a pump-work diagnostic and stress-energy completion for the controller, preserve canonical/phantom sectors consistently through the initial solve and evolution, and then rerun the validation matrix. Only after that source-consistent rerun should confinement, world-tube objectives, and non-dispersing matter models be optimized further.

VII. CONCLUSION

We presented an automatic matter-first discovery pipeline: MAP-Elites proposes scalar configurations, the constraints are solved, the spacetime is evolved, and HQ 4D null geodesics ask whether light arrives earlier than in flat space. Candidate 118, a driven breathing five-lump shell, yields $f_{\text{geo}} = 17.38\%$ in the headline run—light 17% earlier—and 17.27–19.63% across five completed validation runs. The shortcut passes ray and null-Hamiltonian quality gates and is insensitive to the tested domain sizes. These numbers robustly characterize the audited driven implementation, but the unmatched initial/evolution sign convention and non-conservative PD

source preclude a claim of a self-consistent physical warp solution. Candidate 118 is therefore a reproducible numerical precursor requiring a source-consistent rerun, not a continuum-extrapolated or functional warp drive.

VIII. CODE AND DATA AVAILABILITY

The project repository contains the campaign configuration and archive, candidate-118 frozen provenance, the five completed validation summaries, evolving geodesic artifacts, full constraint histories, and post-processed curvature modes. Figure files live in `research/neuralspacetime/figures/`; the article directory holds the reduced numerical tables used by TikZ/PGFPlots.

IX. ACKNOWLEDGEMENTS

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